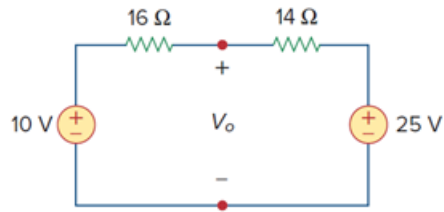


Problem

Determine V_o in the circuit in Fig. 2.80.

Figure 2.80:



Step-by-step solution

Step 1 of 3

Refer to Figure 2.80 in the textbook.

Redraw the circuit diagram showing with loop currents are shown in Figure 1.

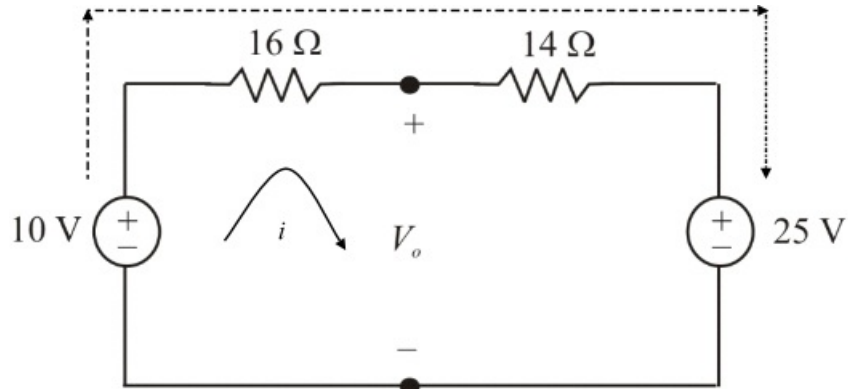


Figure 1

Step 2 of 3

Apply Kirchhoff's voltage law to outer loop in Figure 1.

$$-10 + 16i + 14i + 25 = 0$$

$$30i = -15$$

$$i = -0.5 \text{ A}$$

Step 3 of 3

Apply Kirchhoff's voltage law to first loop in Figure 1.

$$-10 + 16i + V_o = 0$$

Substitute -0.5 A for i .

$$-10 + (16)(-0.5) + V_o = 0$$

$$-10 - 8 + V_o = 0$$

$$V_o = 18 \text{ V}$$

Therefore, the voltage V_o is 18 V.

